

KOLKATA BOSE INTL, India

WMO: 428090

Lat: 22.655N

Lon: 88.447E

Elev: 5

StdP: 101.27

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 11.2	(c) 12.2	(d) 6.9	(e) 6.2	(f) 22.2	(g) 8.1	(h) 6.7	(i) 20.8	(j) 5.4	(k) 22.3	(l) 5.0	(m) 22.1	(n) 0.6	(o) 10	(p) 0.336

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	9.5	37.9	27.3	36.9	27.5	35.9	27.2	29.7	34.7	29.3	34.1	28.9	33.4	3.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.8	25.3	32.5	28.2	24.4	31.9	27.9	24.0	31.7	98.8	34.8	96.7	34.1	94.6	33.5	32.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
6.8	5.9	5.2	DB	9.4	40.3	1.0	1.4	8.7	41.3	8.1	42.2	7.6	43.0	6.9	44.1	
			WB	8.6	30.9	0.8	0.8	8.0	31.4	7.5	31.9	7.1	32.3	6.5	32.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.9	19.1	23.1	27.7	30.4	31.0	30.5	29.6	29.5	29.4	28.1	24.6	20.3
	DBStd	4.41	2.01	2.57	2.40	2.11	1.97	1.77	1.34	1.27	1.40	1.70	1.88	2.02
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	21	15	1	0	0	0	0	0	0	0	0	0	6
	CDD10.0	6185	281	366	547	613	650	615	607	604	583	561	439	319
	CDD18.3	3164	37	133	289	363	392	365	348	346	333	302	189	67
	CDH23.3	37448	250	1096	3228	4793	5443	4860	4326	4251	4020	3303	1484	394
Wind	WSAvg	1.9	1.2	1.3	1.9	3.0	3.1	2.6	2.3	2.1	1.8	1.3	1.0	1.1
	PrecAvg	1675	13	23	28	59	131	295	335	337	286	126	27	8
	PrecMax	2512	128	86	126	206	347	1042	668	620	763	361	297	101
	PrecMin	1082	0	0	0	0	14	78	88	126	39	0	0	0
	PrecStd	344	22	24	30	52	77	181	139	130	162	82	56	21
	DB	29.0	34.1	37.1	39.8	39.3	38.2	35.9	35.1	35.2	34.9	32.2	29.1	29.1
	MCWB	20.1	21.3	23.2	25.2	27.8	29.2	28.7	28.2	27.5	26.7	23.5	20.6	20.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	27.1	32.0	35.2	37.9	37.8	36.9	34.3	34.1	34.2	33.8	31.2	27.9	27.9
	2% MCWB	18.7	21.3	23.4	26.1	28.4	29.1	28.3	28.1	27.7	26.7	23.0	19.9	19.9
	5% DB	26.0	30.3	34.2	36.8	36.8	35.8	33.2	33.1	33.4	32.9	30.2	26.9	26.9
	5% MCWB	17.8	20.5	23.3	26.4	28.2	28.8	28.0	27.9	27.5	26.2	22.5	19.2	19.2
	10% DB	24.8	29.0	33.1	35.2	35.8	34.4	32.2	32.1	32.8	31.9	29.2	25.9	25.9
	10% MCWB	17.4	20.0	23.1	26.4	27.9	28.4	27.7	27.6	27.4	25.6	21.9	18.6	18.6
	WB	22.2	24.6	27.4	29.8	30.6	30.5	29.7	29.5	29.2	28.6	26.0	22.5	22.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% MCDB	26.4	29.1	32.8	35.9	36.0	35.8	33.6	33.0	33.1	32.2	29.1	26.2	26.2
	2% WB	20.5	23.6	26.6	28.9	29.7	29.7	29.1	28.8	28.6	27.9	25.2	21.2	21.2
	2% MCDB	24.6	27.8	31.5	35.0	35.2	34.7	32.7	32.1	32.2	31.3	28.5	25.6	25.6
	5% WB	19.3	22.8	26.0	28.2	29.2	29.3	28.6	28.4	28.2	27.4	24.4	20.4	20.4
	5% MCDB	23.7	26.9	30.6	33.7	34.6	33.9	31.9	31.5	31.5	30.6	27.9	25.0	25.0
	10% WB	18.3	21.8	25.3	27.5	28.7	28.8	28.2	28.0	27.7	26.8	23.6	19.6	19.6
	10% MCDB	22.9	26.4	29.6	32.6	33.7	33.1	31.1	30.8	30.8	29.9	27.3	24.1	24.1
Mean Daily Temperature Range	MDBR	11.6	11.7	11.1	10.2	9.5	7.1	5.7	5.5	6.1	7.5	9.8	11.1	11.1
	5% DB MCDBR	12.8	12.7	11.5	11.1	10.2	9.2	7.1	6.8	7.4	8.3	10.0	11.5	11.5
	5% DB MCWBR	5.4	4.4	4.4	4.3	4.1	4.0	3.1	2.8	2.8	3.2	3.6	4.8	4.8
	5% WB MCDBR	11.1	10.7	10.0	9.9	9.6	8.2	6.4	6.1	6.7	7.2	8.1	10.5	10.5
	5% WB MCWBR	5.5	4.8	4.7	4.5	4.7	4.1	3.3	3.1	3.0	3.3	3.8	5.1	5.1
	taub	0.601	0.564	0.602	0.642	0.690	0.689	0.595	0.565	0.573	0.595	0.635	0.646	0.646
	taud	1.709	1.822	1.784	1.758	1.702	1.714	1.924	2.019	1.980	1.872	1.711	1.667	1.667
Clear-Sky Solar Irradiance	Ebn at Noon	657	730	726	706	669	665	730	754	739	696	626	600	600
	Edh at Noon	217	205	222	231	243	238	193	176	180	192	214	219	219
	RadAvg	3.76	4.72	5.51	6.00	5.71	4.46	4.05	4.16	4.27	4.26	3.99	3.50	3.50
All-Sky Solar Radiation	RadStd	0.23	0.27	0.25	0.29	0.37	0.35	0.33	0.25	0.22	0.32	0.22	0.30	0.30

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(a) Station Only	N/A	(e) N/A	(f) N/A	(g) +0.45	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) +94
(a) Regional (1 neighbor)	N/A	(e) N/A	(f) N/A	(g) +0.45	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) +94

Nomenclature: See separate page